

CPU OVERCLOCKING GUIDE

Overclocking your CPU is a bit complicated but the rewards are worth the effort.

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The easiest way to get more performance from your system is by overclocking the CPU. All consumer grade CPUs have a built-in headroom which is automatically accessed should the need arise by means of that CPU's Turbo frequency. The extent of Turbo on your CPU is determined by your motherboard and your CPU SKU. Lower-end SKUs will not have a high turbo frequency and in the case of AMD Ryzen CPUs, the turbo (XFR as AMD calls it) also depends on the motherboard. If you get the X370/X470 board, then you can get the most out of your CPU as long as the cooling is sufficient. Capping CPU abilities on the basis of the motherboard chipset isn't a new concept so keep that in mind when getting your next system.

Before we get our hands dirty and burn a few transistors off the motherboard, we'd like to make it clear that overclocking guides aren't universally applicable. With each new generation of CPUs, there are architectural and software changes that can invalidate certain steps or require additional steps to be performed. For example, when Skylake was introduced, the DMI bus was unhinged from the BCLK (base clock) allowing simpler overclocking. So always check the respective website to understand

what granular changes have been introduced. Most of these will not be mentioned in manuals. As mentioned previously, you cannot get a decent overclock with just any PC. That being said, let's get started.

BEFORE YOU START

The instructions mentioned here are applicable to both Intel and AMD CPUs on desktop PCs and laptops. Overclocking your CPU will void its warranty but manufacturers do provide overclocking warranty for CPUs. The additional cost depends on the CPU SKU. For a Core i3 CPU it will be \$30 whereas for a top-of-the-line Core i9 it will be \$150. For AMD, the warranty is void the moment you use an aftermarket cooler with your Ryzen CPU and there are no additional warranty plans. For most mainstream Core i7 CPUs it will be \$50. Here's what you need to worry about before overclocking:

The right CPU

While all CPUs can be overclocked, there are limitations baked into the processors that limit the amount of overclock that you can achieve. The first thing to note is if the CPU is unlocked or locked. With Intel CPUs, this is indicated by the 'K' suffix at the end of the model number i.e. Core i7-8700K or Core i9-9900K. With AMD, all CPUs are unlocked. When we say

'locked', we are referring to the CPU multiplier. The processor frequency is calculated by multiplying the base clock with the CPU multiplier so a 3.9 GHz frequency is achieved by a CPU multiplier set to x39 and the Base Clock set to 100 MHz.

Then there's the fact that not all CPUs come out as equals. Certain defects creep into the CPU silicon during the manufacturing process and entire batches can underperform. Manufacturers use a process called binning to explore the potential of a CPU and then they're labelled according to how well they perform. So a Core i7-8700K is a better binned CPU than the Core i7-8700. However, even within the same SKUs, there can be inconsistencies so a certain batch of Core i7-8700K CPUs manufactured on day 1 will not necessarily match those that are made on day 2. Enthusiasts buy CPUs and report how well a certain batch performs. These better binned CPUs can command quite the price premium in enthusiast circuits.

The right motherboard

There's no point getting a really good CPU without investing in a motherboard that can extract the most from that CPU. A good overclocking motherboard boasts of three features - excellent quality components, a good VRM that provides stable voltages and a BIOS that allows granular modifications to all key parameters. Overclocking motherboards are often